REVISION 2021 - PROGRAM CORE - SEMESTER 3

Fundamentals of Electrical Engineering

A professional diploma engineering handbook for Mechanical Engineering and Manufacturing Technology students. It converts the official syllabus into a practical learning guide with circuit laws, AC fundamentals, motors, starters, transformers, heating, welding and electronics applications.

COURSE CODE
3024

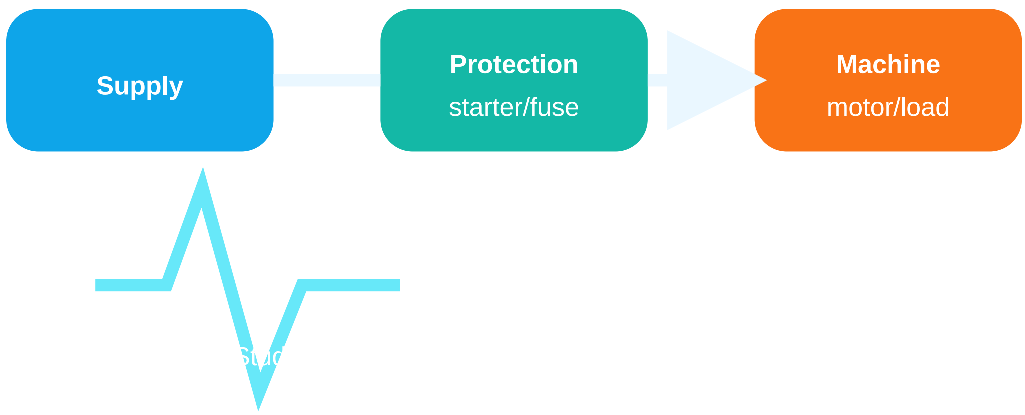
CREDITS
4

PERIODS
60

CATEGORY
Program Core

> Mechanical / Manufacturing students-□□□□ Electrical Engineering □□□□□□□□□□□□ machine, motor, starter, heating, welding, drive, sensor, rectifier □□□□□□ practical □□□ □□□□□□□□□□□□□□ □□□□□□□□□□. Formula □□□□□□ memorize □□□□□□□□□; circuit path, unit, application □□□□□□ □□□□ □□□□□□□□.

Engineering signal flow



Course Overview

Officially confirmed from the supplied syllabus: program, code, title, semester, credits, period structure, objectives, prerequisites, COs, CO-PO mapping, module outcomes, contents, references and online resources.

Why this subject matters

Mechanical engineers work with electrically powered machines every day: pumps, compressors, conveyors, lifts, welding plants, furnaces, machine tools and EV charging systems. A weak understanding of current, voltage, power, motor starters and electronics leads to unsafe troubleshooting and wrong maintenance decisions.

How to study

1. Start with units: V, A, Ohm, W, Wh and kWh.
2. For every circuit, mark current path before applying formula.
3. For every machine, draw supply, protection, starter, motor and load.
4. For every electronics topic, draw input, device, output and load.

Source information table

Item	Official value
Programme	Diploma in Mechanical Engineering / Manufacturing Technology
Course title	Fundamentals of Electrical Engineering
Course code	3024
Semester	3 / 3
Credits	4
Periods	4 per week, 60 per semester
Prerequisites	Applied Physics I & II and Mathematics I & II
CO-PO mapping	CO1 strongly mapped to PO1; CO2, CO3 and CO4 moderately mapped to PO1

Table of Contents

Redesigned Syllabus Roadmap

This roadmap converts the official table into what a student must draw, calculate, explain and apply in field work.

CO1 - 15 h

Electrical circuits, DC laws, AC fundamentals, power and energy.

Draw: series/parallel circuits, AC waveform, star/delta. Calculate: equivalent resistance, KCL/KVL, RMS, power, energy bill.

Common mistake: mixing W and Wh, line and phase values.

CO2 - 15 h

DC motor, three-phase induction motor, single-phase motor and starters.

Draw: motor principle, DOL, star-delta and 3-point starter. Explain: starting current, rotating field and applications.

Common mistake: saying starter only switches ON/OFF.

CO3 - 14 h

Transformer, auto transformer, welding transformer, electric heating and furnaces.

Draw: transformer core, induction furnace, arc furnace. Calculate: EMF equation and transformation ratio.

Common mistake: confusing induction heating with resistance heating.

CO4 - 14 h

Electronics components, rectifiers, BJT switch, SCR circuits, drives and EV charging block.

Draw: diode rectifier, BJT switch, SCR, chopper, drive block and EV charger block.

Common mistake: not showing load and supply direction.

CO1 - APPLYING - 15 HOURS

Module 1: Electrical Circuits, AC Fundamentals, Power and Energy

Solve simple electrical circuits using circuit parameters and basic laws.

Core explanation

Current, EMF and resistance

Current is the flow of electric charge. EMF is the source pressure that pushes charge. Resistance opposes current. For mechanical students, treat voltage like pressure, current like flow and resistance like restriction, but remember electrical quantities must be calculated using proper units.

Series and parallel resistance

In series, same current flows and resistances add. In parallel, same voltage appears and reciprocal conductance adds. Combination problems must be simplified stage by stage, not guessed.

Ohm and Kirchhoff laws

Ohm law links V, I and R for linear resistive circuits. KCL says current entering a junction equals current leaving. KVL says algebraic sum of voltage in a closed loop is zero.

Electromagnetic induction

Faraday law states induced EMF depends on rate of change of flux linkage. Lenz law gives the direction: induced effect opposes the cause. Motors, transformers and generators depend on this.

AC waveform terms

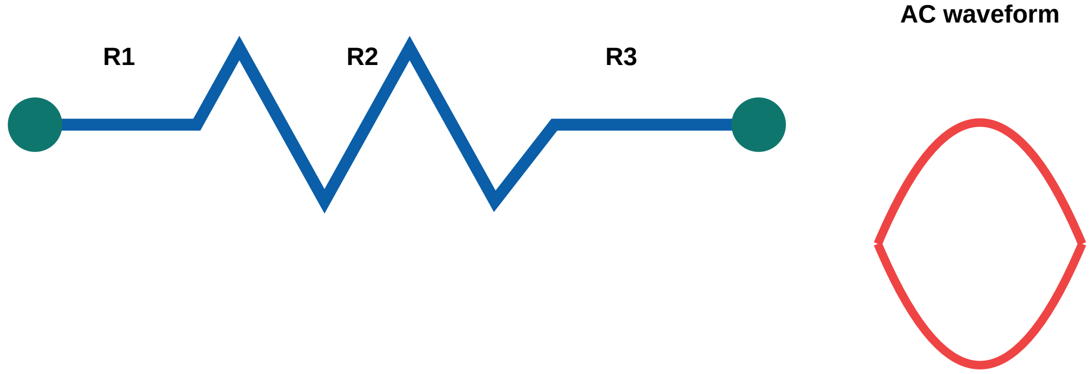
Cycle, frequency, period, amplitude, angular velocity, RMS, average value, form factor and peak factor are used to specify AC quantities. RMS is the heating equivalent DC value.

Three phase star and delta

In star: line voltage equals root 3 times phase voltage and line current equals phase current. In delta: line current equals root 3 times phase current and line voltage equals phase voltage.

KCL/KVL problem □□□□□□□□□□□ □□□□□ current direction assume □□□□□□□. Negative answer □□□□□□□□□ direction opposite □□□□□□□ □□□□□□; formula □□□□□□ □□□□□ □□□□□□□□□□.

Diagram lab



Important formulas

$$V = I R$$

Ohm law. V in volt, I in ampere, R in ohm.

$$R_s = R1 + R2 + ...$$

Series resistance.

$$1/R_p = 1/R1 + 1/R2 + ...$$

Parallel resistance.

$$P = V I = I^2R = V^2/R$$

DC power in watt.

Energy = Power x Time

1 commercial unit = 1 kWh.

$T = 1/f$, $\omega = 2\pi f$

AC period and angular velocity.

Solved example

1. Two resistors 6 ohm and 3 ohm are in parallel. Find equivalent resistance.
2. Use R = R1R2/(R1+R2) for two parallel resistors.
3. R = 6 x 3 / (6+3) = 18/9 = 2 ohm.
4. Exam tip: parallel equivalent must be less than the smallest branch resistance.

Expected questions

- Define current, EMF and resistance.
- Solve a series-parallel resistance problem.
- State and apply Ohm law, KCL and KVL.
- Explain Faraday and Lenz laws.
- Define RMS, average value, form factor and peak factor.
- Calculate monthly electricity bill from kWh.

CO2 - UNDERSTANDING - 15 HOURS

Module 2: Electric Motors and Starters

Explain working principle of electric motors and their mechanical engineering applications.

Textbook chapter

DC motor principle

A current carrying conductor placed in magnetic field experiences force. In a DC motor, armature conductors carry current and field system produces flux. Commutator keeps torque in one direction.

DC motor classification

Based on field connection: series motor, shunt motor and compound motor. Series motor gives high starting torque. Shunt motor gives nearly constant speed.

Three phase induction motor

Three phase supply produces rotating magnetic field in stator. This field induces rotor current and torque. It is rugged, reliable and widely used in pumps, compressors, conveyors and machine tools.

Single phase induction motor

Single phase motor is not self-starting by basic single winding. Auxiliary winding or capacitor produces starting torque. Used in fans, small pumps and domestic machinery.

Starter necessity

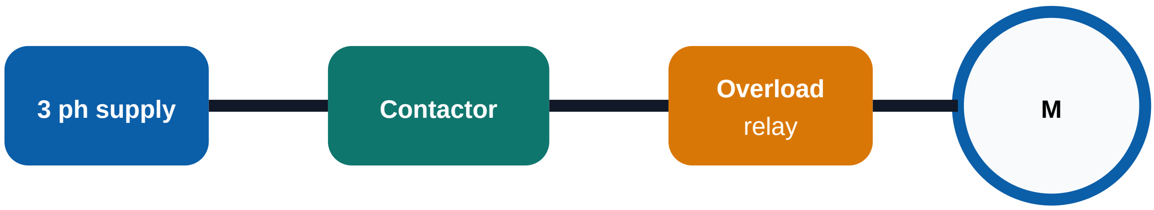
At starting, motor back EMF is low and current can become very high. Starter limits starting current, gives protection and allows safe starting sequence.

DOL and star-delta starters

DOL directly connects motor to supply and is used for smaller motors. Star-delta starter starts in star to reduce starting current, then changes to delta for running.

Starter □□□ switch □□□□□□□□□□. Starting current limit □□□□□□□□□□ overload/no-voltage protection □□□□□□□□ □□□□□□□□□□□□□□□□.

Motor starter diagram



Supply - starter - protection - motor - mechanical load

Applications and troubleshooting

Motor/system	Application	Fault clue	Maintenance check
DC series motor	High starting torque applications	Brush sparking, poor speed control	Brush, commutator, field connection
3 phase induction motor	Pumps, compressor, conveyor, escalator drive systems	Humming, overload trip, single phasing	Voltage balance, bearing, starter, overload setting
Single phase motor	Small machines, fans, pumps	Will not start, capacitor failure	Capacitor, auxiliary winding, centrifugal switch

Expected questions

Explain DC motor principle and classification.

Explain construction and working of three phase induction motor.

Explain single phase induction motor and applications.

Why is starter required?

Explain 3-point DC starter, DOL starter and star-delta starter.

CO3 - UNDERSTANDING - 14 HOURS

Module 3: Transformers, Electric Heating and Welding Equipment

Illustrate electrical equipment used in mechanical manufacturing processes.

Textbook chapter

Transformer principle

A transformer works on mutual induction. AC in primary winding creates changing flux in the core and induces EMF in secondary winding. It changes voltage level without changing frequency.

EMF equation and ratio

Transformer EMF depends on frequency, number of turns and maximum flux. Transformation ratio links secondary and primary voltage or turns.

Auto transformer

Auto transformer uses a single winding with tapping. It is compact and efficient but has no electrical isolation between primary and secondary.

Welding transformer

Welding transformer gives low voltage and high current needed for arc welding. Current control and thermal design are important.

Electric heating

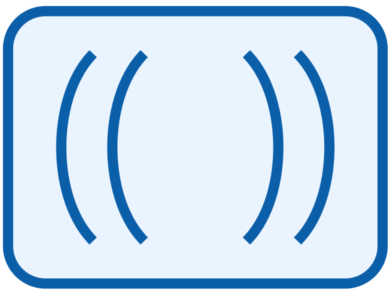
Heat can be produced by resistance, induction, dielectric and arc methods. Manufacturing uses these for furnaces, heat treatment, welding and melting.

Furnaces

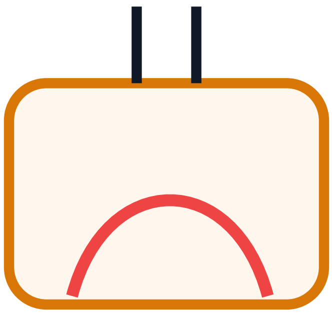
Induction furnace heats conductive material by induced currents. Arc furnace uses electric arc heat and is used for melting metals.

Transformer DC-□ work □□□□□□□□□□. Changing flux □□□□. Induction furnace-□ heat workpiece-□□□□ □□□□□□ induce □□□□□□□□□□ current □□□□□□□□.

Transformer and furnace visuals



Transformer



Arc / induction furnace idea

Important formulas

$$E = 4.44\, f\, N\, \Phi_m$$

Transformer EMF equation.

$$V1/V2 = N1/N2$$

Transformation ratio for ideal transformer.

$$H = I^2 R t$$

Joule heating in resistance heating.

Expected questions

Explain construction and working principle of single-phase transformer.

Write transformer EMF equation and transformation ratio.

Explain auto transformer and welding transformer.

Compare resistance, induction and dielectric heating.

Explain induction furnace and arc furnace.

CO4 - UNDERSTANDING - 14 HOURS

Module 4: Electronics Applications in Mechanical Engineering

Summarize components, rectifiers, BJT/SCR switching, electric drives and EV charging.

Textbook chapter

Active and passive components

Passive components such as resistors and capacitors do not provide power gain. Active devices such as diodes, transistors and SCRs control current and can switch or amplify signals.

PN junction diode and rectifier

A diode conducts mainly in forward bias and blocks reverse bias. Rectifier circuits convert AC to DC. Half-wave and full-wave rectifiers are common basic forms.

BJT as switch

In cutoff, transistor is OFF. In saturation, it acts as ON switch. It can drive relays, lamps and small control loads through proper current limiting and protection.

SCR based circuits

SCR is a controlled rectifier. It turns ON with gate trigger and remains ON until current falls below holding current. Used in controlled rectifiers and power control.

Chopper

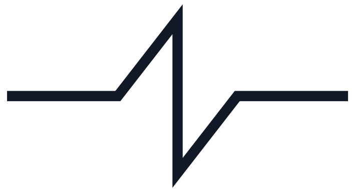
A chopper converts fixed DC to variable DC using high-speed switching. Average output voltage is controlled by duty cycle.

Electric drives and EV charging

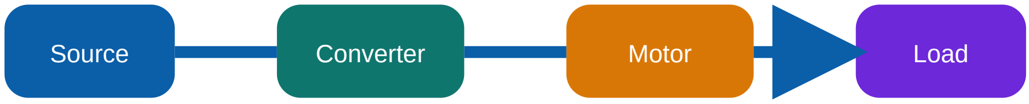
An electric drive includes source, converter, controller, motor and feedback. EV charger blocks include AC input, protection, rectifier/PFC, DC link, DC-DC converter, control and battery management.

> Electronics applications □□□□□□□□□□□□ device symbol, current direction, input/output, load, protection □□□□□□ diagram-□ □□□□□□□□□.

Rectifier and drive block



Diode rectifier idea



Field relevance for mechanical work

Concept	Where seen in workshop / plant	Check before troubleshooting
Rectifier	Battery charger, DC control supply, welding control	Input AC, diode short/open, output DC ripple
BJT switch	Relay driver, sensor interface	Base resistor, coil diode, ground continuity
SCR	Heater control, controlled rectifier, chopper concept	Gate trigger, holding current, heat sink
Drive	Motor speed control, conveyor, pump, escalator/elevator drive systems	Input supply, DC bus, feedback, fault code, earthing

Expected questions

- Distinguish active and passive components.
- Explain PN junction diode and rectifier circuits.
- Explain transistor as a switch.
- Explain SCR working and basic SCR based circuits.
- Draw electric drive block diagram and EV charging block diagram.

Formula / Rule Bank

Every formula below includes where it is used. Use correct units and write substitutions in exams.

Ohm law

$$V = IR$$

Used for DC resistive circuits. Example: 12 V across 6 ohm gives 2 A.

KCL

$$\Sigma I_{in} = \Sigma I_{out}$$

Used at junctions and node problems.

KVL

$$\Sigma V = 0$$

Used in closed loops.

Power

$$P = VI$$

Unit watt. Used for electrical load rating.

Energy

$$E = P \times t$$

kWh is commercial unit.

AC relation

$$V_{rms} = V_m/\sqrt{2}$$

For sinusoidal voltage.

Impedance

$$Z = V/I$$

AC opposition including resistance and reactance.

Power factor

$$PF = \cos\varphi$$

Used in AC power and motor systems.

Transformer

$$E = 4.44\ f\ N\ \Phi_m$$

EMF equation.

Ratio

$$V1/V2 = N1/N2$$

Ideal transformer ratio.

Joule heating

$$H = I^2Rt$$

Resistance heating principle.

Chopper rule

$$V_o = D\ V_s$$

Average output in step-down DC chopper concept.

Solved Examples, Practice and Expected Questions

Solved 1: Ohm law

1. Given R = 10 ohm and I = 2 A.
2. Formula V = IR.
3. V = 2 x 10 = 20 V.
4. Answer: 20 V.

Solved 2: Energy bill

1. A 1 kW heater works for 3 h daily for 30 days.
2. Energy = 1 x 3 x 30 = 90 kWh.
3. If rate is Rs 6 per unit, bill = 90 x 6 = Rs 540.

Solved 3: Transformer ratio

1. Primary 1000 turns, secondary 250 turns, V1 = 230 V.
2. V2 = V1 x N2/N1.
3. V2 = 230 x 250/1000 = 57.5 V.

Solved 4: Parallel resistance

1. R1 = 4 ohm, R2 = 12 ohm.
2. R = R1R2/(R1+R2).
3. R = 48/16 = 3 ohm.

Objective practice

Unit of electrical energy in billing is: A) W B) kWh C) V D) Ohm. Answer: B.

Three phase induction motor works on: A) electrolysis B) rotating magnetic field C) photoelectric effect D) heating only. Answer: B.

A transformer works on: A) mutual induction B) chemical action C) piezoelectricity D) radiation. Answer: A.

BJT in saturation acts as: A) open switch B) ON switch C) capacitor D) fuse. Answer: B.

SCR terminal names are: A) emitter base collector B) anode cathode gate C) drain gate source D) L N E. Answer: B.

Module-wise expected questions

Module	One mark	Short answer	Descriptive/diagram
M1	Define RMS, PF, kWh	Ohm/KCL/KVL problems	AC generation and star/delta relations
M2	Types of motors/starters	Need of starter	3-phase IM, single-phase IM, DOL and star-delta starter
M3	Transformer ratio, heating modes	Auto/welding transformer	Induction and arc furnace
M4	Active/passive, SCR, BJT	Rectifier and transistor switch	Electric drive and EV charging block

Fresh Model Question Paper and Complete Answer Guide

This is a fresh practice paper based on the supplied syllabus. It does not copy official questions.

Model Paper

Course: 3024 Fundamentals of Electrical Engineering
Time: 3 Hours
Maximum Marks: 100

Part A - Short Answer
1. Define current, EMF and resistance.
2. State Ohm law.
3. Define RMS value and power factor.
4. State Faraday law and Lenz law.
5. List types of DC motors.

6. Why is a motor starter required?
7. State transformer EMF equation.
8. List methods of electric heating.
9. Distinguish active and passive components.
10. Name the terminals of SCR.

Part B - Problem / Explanation
11. Solve a series-parallel resistance circuit.
12. Explain Kirchhoff laws with example.
13. Explain AC waveform terms with diagram.
14. Explain working principle of DC motor.
15. Explain three phase induction motor construction and working.
16. Explain DOL and star-delta starters.
17. Explain single phase transformer with EMF equation.
18. Explain welding transformer and electric heating methods.
19. Explain diode rectifier circuit.
20. Explain BJT as switch and SCR working.

Part C - Descriptive / Application
21. Calculate monthly electricity bill for given loads and tariff.
22. Draw and explain induction furnace and arc furnace.
23. Draw electric drive block diagram and explain elements.
24. Draw EV charging system block diagram and explain flow.
25. Write safety and troubleshooting checks for motor starter panel.

Complete model answer guide

Current is charge flow; EMF is source voltage; resistance opposes current.

V = IR. Mention units.

RMS is DC equivalent heating value; PF = cos phi.

Induced EMF depends on rate of change of flux; direction opposes cause.

Series, shunt and compound DC motors.

Starter limits starting current and gives protection.

E = 4.44 f N Phi max.

Resistance, induction, dielectric and arc heating.

Passive components do not give power gain; active devices control current or amplify/switch.

SCR terminals: anode, cathode and gate.

For long answers: draw diagram first, then write working, formulas, applications, safety and common faults.

References

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